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Question 1

Correct

Marked out of 1.00

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|           |                                  |
|-----------|----------------------------------|
| Status    | Finished                         |
| Started   | Sunday, 12 January 2025, 7:31 PM |
| Completed | Sunday, 12 January 2025, 7:38 PM |
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A binary number is a combination of 1s and 0s. Its  $n^{\text{th}}$  least significant digit is the  $n^{\text{th}}$  digit starting from the right starting with 1. Given a decimal number, convert it to binary and determine the value of the  $4^{\text{th}}$  least significant digit.

Example

number = 23

- Convert the decimal number 23 to binary number:  $23^{10} = 2^4 + 2^2 + 2^1 + 2^0 = (10111)_2$ .
- The value of the  $4^{\text{th}}$  index from the right in the binary representation is 0.

Function Description

Complete the function fourthBit in the editor below.

fourthBit has the following parameter(s):

int number: a decimal integer

Returns:

int: an integer 0 or 1 matching the 4th least significant digit in the binary representation of number.

Constraints

$0 \leq \text{number} < 2^{31}$

Input Format for Custom Testing

Input from stdin will be processed as follows and passed to the function.

The only line contains an integer, number.

Sample Case 0

Sample Input 0

STDIN   Function  
-----  
32   →   number = 32

Sample Output 0

0

Explanation 0

- Convert the decimal number 32 to binary number:  $32_{10} = (100000)_2$ .
- The value of the 4th index from the right in the binary representation is 0.

Sample Case 1

Sample Input 1

STDIN   Function  
-----  
77   →   number = 77

Sample Output 1

1

Explanation 1

- Convert the decimal number 77 to binary number:  $77_{10} = (1001101)_2$ .
- The value of the 4th index from the right in the binary representation is 1.

**Answer:** (penalty regime: 0 %)

Reset answer

```
1  /*
2  * Complete the 'fourthBit' function below.
3  *
4  * The function is expected to return an INTEGER.
5  * The function accepts INTEGER number as parameter.
6  */
7
8  int fourthBit(int number)
9  {
10     return(number >> 3) & 1;
11 }
```

|   | Test                        | Expected | Got |   |
|---|-----------------------------|----------|-----|---|
| ✓ | printf("%d", fourthBit(32)) | 0        | 0   | ✓ |
| ✓ | printf("%d", fourthBit(77)) | 1        | 1   | ✓ |

Passed all tests! ✓

Question **2**

Correct

Marked out of 1.00

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Determine the factors of a number (i.e., all positive integer values that evenly divide into a number) and then return the  $p^{\text{th}}$  element of the list, sorted ascending. If there is no  $p^{\text{th}}$  element, return 0.

**Example**

$n = 20$   
 $p = 3$

The factors of 20 in ascending order are {1, 2, 4, 5, 10, 20}. Using 1-based indexing, if  $p = 3$ , then 4 is returned. If  $p > 6$ , 0 would be returned.

**Function Description**

Complete the function `pthFactor` in the editor below.

`pthFactor` has the following parameter(s):

`int n`: the integer whose factors are to be found

`int p`: the index of the factor to be returned

Returns:

`int`: the long integer value of the  $p^{\text{th}}$  integer factor of  $n$  or, if there is no factor at that index, then 0 is returned

**Constraints**

$$1 \leq n \leq 10^{15}$$

$$1 \leq p \leq 10^9$$

Input Format for Custom Testing

Input from stdin will be processed as follows and passed to the function.

The first line contains an integer  $n$ , the number to factor.

The second line contains an integer  $p$ , the 1-based index of the factor to return.

**Sample Case 0**

**Sample Input 0**

|       |            |
|-------|------------|
| STDIN | Function   |
| ----- | -----      |
| 10    | → $n = 10$ |
| 3     | → $p = 3$  |

**Sample Output 0**

Explanation 0

Factoring  $n = 10$  results in  $\{1, 2, 5, 10\}$ . Return the  $p = 3^{\text{rd}}$  factor, 5, as the answer.

Sample Case 1

Sample Input 1

STDIN    Function

10    →     $n = 10$

5    →     $p = 5$

Sample Output 1

0

Explanation 1

Factoring  $n = 10$  results in  $\{1, 2, 5, 10\}$ . There are only 4 factors and  $p = 5$ , therefore 0 is returned as the answer.

Sample Case 2

Sample Input 2

STDIN    Function

1    →     $n = 1$

1    →     $p = 1$

Sample Output 2

1

Explanation 2

Factoring  $n = 1$  results in  $\{1\}$ . The  $p = 1^{\text{st}}$  factor of 1 is returned as the answer.

Answer: (penalty regime: 0 %)

Reset answer

```
1 1 /*
2  * Complete the 'pthFactor' function below.
3  *
4  * The function is expected to return a LONG_INTEGER.
5  * The function accepts following parameters:
6  * 1. LONG_INTEGER n
7  * 2. LONG_INTEGER p
8  */
9
10 long pthFactor(long n, long p)
11 {
12     long j = 0;
13
14     for(long i = 1; i <= n; i++){
15         if(n % i == 0){
16             j++;
17             if(j == p){
18                 return i;
19             }
20         }
21     }
22     return 0;
23 }
```

|   | Test                            | Expected | Got |   |
|---|---------------------------------|----------|-----|---|
| ✓ | printf("%ld", pthFactor(10, 3)) | 5        | 5   | ✓ |
| ✓ | printf("%ld", pthFactor(10, 5)) | 0        | 0   | ✓ |
| ✓ | printf("%ld", pthFactor(1, 1))  | 1        | 1   | ✓ |

Passed all tests! ✓